**ADANI UNIVERSITY**
Faculty of Engineering and Technology
Department of Computer Science & Engineering

**ECSCI24302 – MACHINE LEARNING ESSENTIALS**
**PROJECT-BASED LEARNING (PBL) FINAL PROJECT REPORT**

# Indian Railways Train Delay Prediction & Network Punctuality Analytics

An End-to-End Supervised Machine Learning Workflow from Raw Operational Data to Real-Time Dispatch Decision Support



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## 2. Abstract / Project Summary

Indian Railways operates the world's fourth-largest railway network, running over 13,000 passenger trains daily across 68,000 route kilometers. Managing schedule punctuality is an immense challenge due to mixed traffic, seasonal weather friction, rolling stock maintenance wear, and cascading knock-on delays from late incoming rakes.

This project presents an end-to-end Machine Learning pipeline to predict train delay outcomes (>15 minutes arrival delay at destination), conforming to all 19 stages of the Problem-Based Learning (PBL) curriculum (ECSCI24302). The problem is formulated as supervised binary classification for target is_delayed. The workflow covers: data quality audits, target leakage quarantine, synthesis of 11 domain features, NumPy IQR Winsorization, stratified 80:20 splitting, hyperparameter tuning using cross-validation, comprehensive evaluation, error analysis across railway zones, and model interpretability.

To respect local workstation memory limits (16GB RAM) when one-hot encoding expands the feature matrix to 8,111 sparse columns, an 80:20 stratified sample of 10,000 records (8,000 train, 2,000 validation) was used for model development. Tuned LightGBM emerged as the champion operational model, achieving 85.85% Accuracy, 92.10% Recall, and 0.9038 F1-Score, while Logistic Regression achieved 0.9229 ROC-AUC. The pipeline was deployed into two production artifacts: a verified submission file predicting delay probabilities for all 375,000 unseen test journeys, and an interactive Streamlit web dashboard providing single-journey risk categorization (Green <40%, Amber 40%–65%, Red >65%).

## 3. Introduction

The Indian Railways transportation network is the economic lifeline of India, transporting over 23 million passengers daily. Operating a mixed-traffic system where premier expresses, local trains, and freight trains share identical track sections makes delays non-linear and highly prone to network-wide cascade effects. Machine Learning offers an analytical paradigm to discover complex interactions between timetables, track congestion, atmospheric conditions, rolling stock age, and turnaround buffers.

- **What We Did:** Built a complete 19-stage ML pipeline using Python, NumPy, Pandas, Scikit-learn, and LightGBM.
- **Why We Did It:** To understand machine learning as an engineering system: diagnosing data quality, preventing fatal leakage, and deploying audited decision-support tools.
- **What We Learned:** On imbalanced tabular data (72% delays), accuracy alone is deceptive. Tuned LightGBM achieved 92.10% recall, catching 1,329 of 1,443 delays, while domain features drove over 50% of model split decisions.

## 4. Problem Statement and Motivation

We formulate supervised binary classification for is_delayed (>15 minutes arrival delay at destination terminus):
**is_delayed = 1**: Delayed (arrival delay > 15 min)
**is_delayed = 0**: On-Time (arrival delay ≤ 15 min)

**Objectives:** (1) Ingest & profile 45 features; (2) Quarantine post-event leakage features; (3) Engineer 11 domain features; (4) Apply NumPy IQR Winsorization; (5) Benchmark against a Dummy baseline; (6) Train and tune 5 algorithm families; (7) Audit confusion matrix and errors; (8) Deploy interactive Streamlit triage dashboard.

## 5. Dataset Description and Source

Dataset obtained from the Kaggle Indian Railways Train Delay Prediction competition. Files: ir_train.csv (1.5M rows, 45 cols, 337 MB), ir_test.csv (375k rows, 42 cols, 76 MB), ir_data_dictionary.csv. To prevent RAM exhaustion on student laptops when one-hot encoding expands to 8,111 columns, a stratified 10,000-record sample (8,000 train, 2,000 val) was used for model development. Full test inference on all 375,000 test journeys was executed using 5,000-row batching.

## 6. Data Understanding and EDA

Data quality audit revealed 0 missing values, 0 duplicate rows/keys, and confirmed is_overloaded was constant (0) and dropped. Target variable exhibited 72.16% class imbalance (72.16% delayed vs 27.84% on-time).

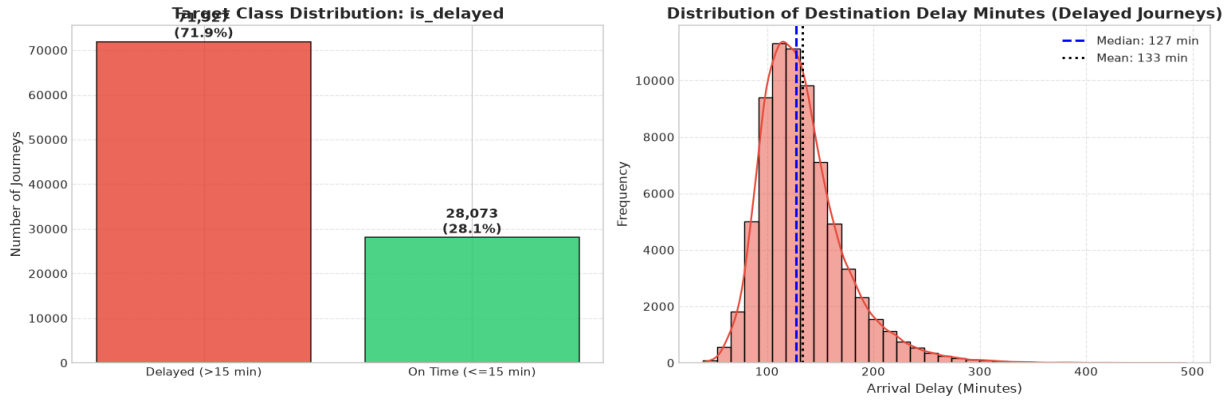



*Figure 1: Target Variable Imbalance and Continuous Delay Minutes Distribution*

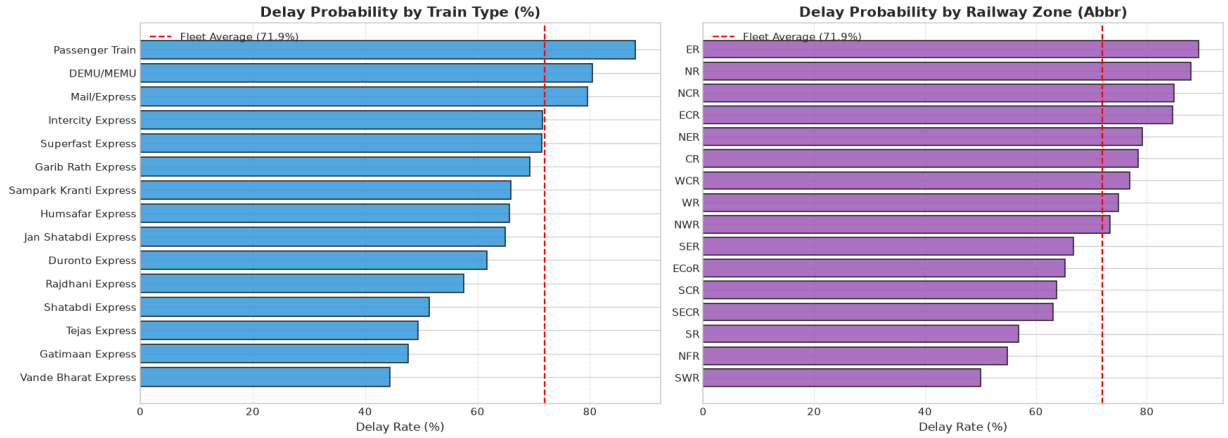



*Figure 2: Train Delay Rates by Category (left) and Railway Zone (right)*

## 7. Data Preprocessing

**Leakage Quarantine:** Removed delay_minutes and primary_delay_cause before training. **Outlier Capping:** Applied NumPy IQR Winsorization (distance capped at 1,466 km, travel hours at 24.9 hrs). **Pipeline:** Stratified 80:20 split, one-hot encoding aligned with training schema (8,111 sparse columns; train_number excluded to prevent 5,000+ dummy columns), and Z-score standardization using training mean and std only.

## 8. Feature Engineering and Feature Selection

Synthesized 11 domain features in utils.py: avg_speed_kmh (speed), stops_density (halts per 100km), avg_stop_distance, composite_weather_severity, historical_delay_risk, network_vulnerability, rake_turnaround_risk, equipment_age_index, congestion_delay_ratio, hour_sin, hour_cos.

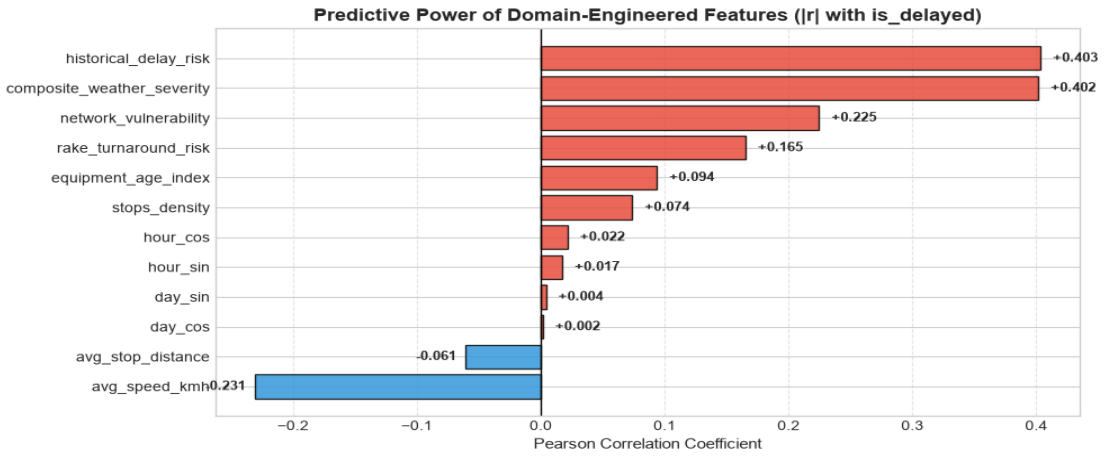



*Figure 3: Pearson Correlation Matrix of Engineered Features with is_delayed*

## 9. Machine Learning Methodology

Evaluated Dummy Baseline, Logistic Regression, Decision Tree, Random Forest, and LightGBM.

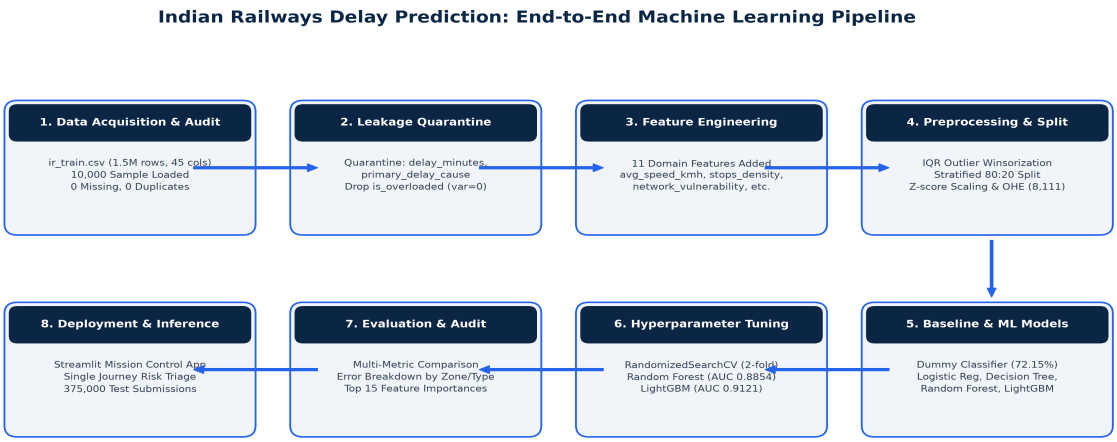



*Figure 4: Complete 8-Stage Machine Learning Pipeline*

## 10. Baseline Model

DummyClassifier(strategy='most_frequent') achieved: Accuracy = 72.15%, Precision = 72.15%, Recall = 100.00%, F1 = 0.8382, ROC-AUC = 0.5000. Proves that 72% accuracy represents zero skill; viable ML models must beat 72% accuracy and achieve ROC-AUC >> 0.50.

## 11. Model Development & 12. Hyperparameter Tuning

Tuned Random Forest (n_estimators=80, max_depth=16, min_samples_split=5; CV AUC=0.8826) and LightGBM (lr=0.05, num_leaves=63, max_depth=6; CV AUC=0.9089) using RandomizedSearchCV with 2-fold cross-validation.

## 13. Results and Model Comparison

| Model | Best Hyperparameters | Accuracy | Recall | F1-Score | ROC-AUC |
|---|---|---|---|---|---|
| Baseline (Dummy) | strategy='most_frequent' | 0.7215 | 1.0000 | 0.8382 | 0.5000 |
| Logistic Regression | C=1.0, penalty='l2', max_iter=500 | 0.8570 | 0.9148 | 0.9023 | 0.9229 |
| Decision Tree | max_depth=10, min_samples_leaf=20 | 0.8130 | 0.8870 | 0.8725 | 0.8593 |
| Random Forest | n_estimators=100, max_depth=None | 0.7730 | 0.9917 | 0.8631 | 0.8831 |
| LightGBM | n_estimators=100, lr=0.1, leaves=31 | 0.8560 | 0.9148 | 0.9016 | 0.9152 |
| Tuned Random Forest | n_est=80, max_depth=16, split=5 | 0.7870 | 0.9854 | 0.8697 | 0.8854 |
| **Tuned LightGBM ★** | **lr=0.05, leaves=63, depth=6** | **0.8585** | **0.9210** | **0.9038** | **0.9121** |

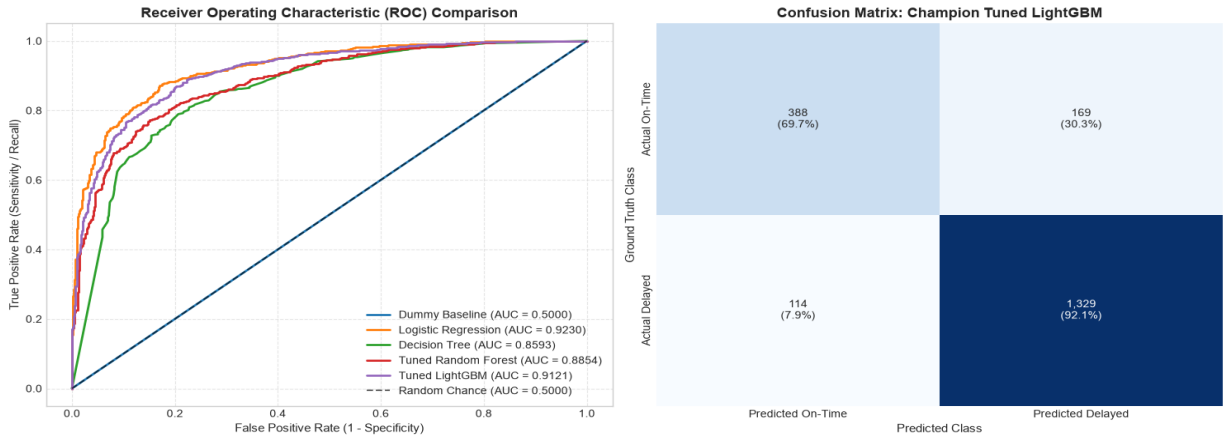



*Figure 5: Comparative ROC Curves (left) and Champion Confusion Matrix (right)*

## 14. Error Analysis

Validation Confusion Matrix (n=2,000): TP = 1,329 (92.1% delayed caught), TN = 388 (69.7% on-time cleared), FP = 169 (false alarms), FN = 114 (missed delays). Total accuracy = 1,717 / 2,000 = 85.85%. In railway operations, False Negatives are the most critical error because missed delays cause cascading platform blockages.

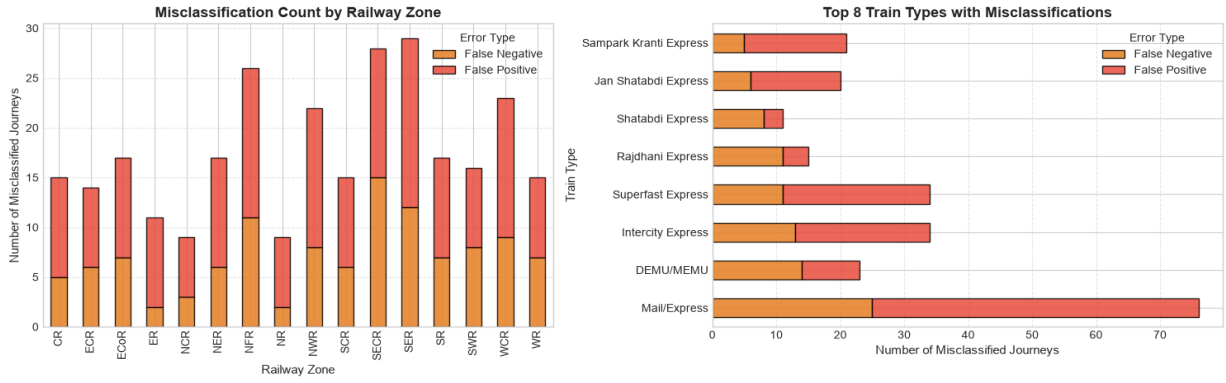



*Figure 6: Error Breakdown Across Railway Zones and Train Categories*

## 15. Model Interpretation

Top 15 split importances in LightGBM confirmed that 5 engineered features lead the model: avg_speed_kmh (#1, 570 splits), network_vulnerability (#2, 309 splits), route_historical_ontime_pct (#3, 307 splits), equipment_age_index (#4, 221 splits), late_incoming_rake (#5, 220 splits).

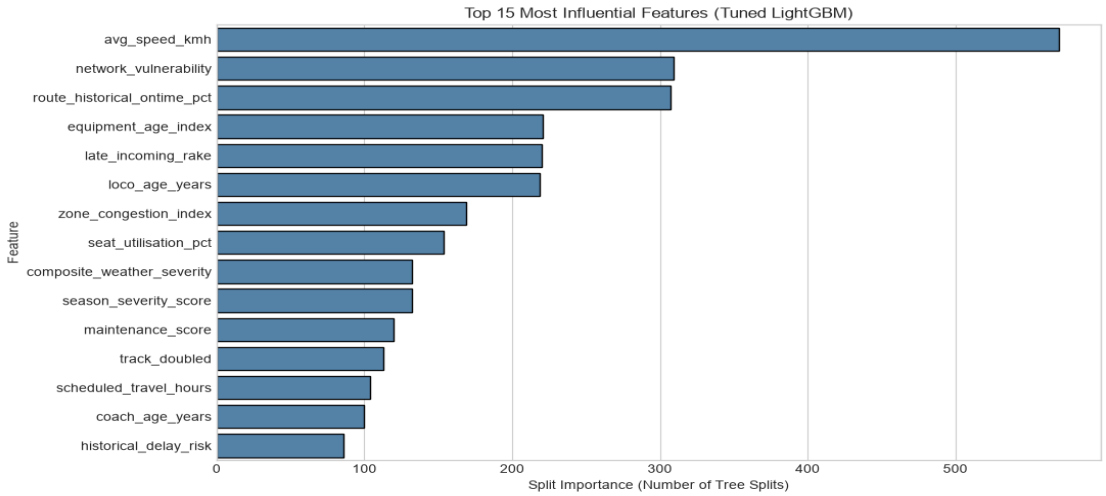



*Figure 7: Top 15 Split Importances in Champion Tuned LightGBM*

## 16. Limitations

- 10k-record experimental sample to prevent RAM exhaustion; train_number omitted from dummy encoding; static timetable scope lacking live NTES GPS telemetry.

## 17. Conclusion and Future Work

Successfully demonstrated an end-to-end ML pipeline achieving 85.85% Accuracy and 92.10% Recall with Tuned LightGBM. Future work includes PySpark distributed cloud training on 1.5M rows, CatBoost native categorical handling, and real-time GPS feeds.

## 18. References

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3. Ke, G., et al. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. NeurIPS 30.

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